EXECUTIVE CONSULTING REPORT — ENGLISH VERSION

# FitBrand Co.: Retention, Unit Economics and Capital Reallocation

*D2C Acquisition Diagnostics and Action Plan by Channel — Python, SQL, and Power BI pipeline*

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## 1. Executive Summary

FitBrand Co. has expanded its customer acquisition through social media channels and its brand ambassador program; however, net margin growth is no longer keeping pace with top-line sales. The core challenge lies in customer retention rather than acquisition volume. Within the first 90 days, FitBrand Co. experiences a steep decline in initial cohort retention, forcing the business to offset lost volume through increasingly expensive paid ad spend. The channel that appears most cost-effective according to media performance reports—TikTok—yields the lowest retention rate: a 21.79% cumulative repeat purchase rate compared to 54.83% for Ambassador.

This analysis is based on 1,807 transactions across 1,000 unique customers (~$71,175 USD in total sales). Average LTV for Ambassador ($84.29 USD) is 55% higher than TikTok ($54.32 USD). Using a benchmark industry gross margin of 65% and simulated channel-level CACs ($8.00 USD commission for Ambassador, $15.00 USD paid media CAC for TikTok), net profit per customer for Ambassador ($46.79 USD) more than doubles TikTok ($20.31 USD). TikTok is not an unprofitable channel per se, but rather a channel that drives volume at the expense of unit margin—a cost structure that deteriorates further as real CAC rises.

Strategic recommendations focus on capital reallocation and activating retention levers rather than merely optimizing ad creatives: reallocate 30% of TikTok ad spend into the Ambassador commission structure, launch product-segmented post-purchase educational retargeting, implement checkout cross-sell bundles to raise initial AOV, and deploy a Subscribe & Save subscription model alongside a points-based loyalty program to secure the second replenishment order. None of these four initiatives require incremental budget: they represent a capital redistribution leveraging existing technology infrastructure.

## 2. Business Context and Unit Economics

### 2.1 The Problem Is Retention, Not Acquisition

Channel-level performance data is conclusive. Ambassador retains 54.83% of customers after their initial purchase; Organic retains 52.94%; Instagram retains 30.48%; while TikTok retains only 21.79%

The core business hypothesis indicates that TikTok-driven traffic stems from impulse buys or low-commitment introductory offers, hindering brand loyalty without a dedicated post-purchase re-engagement strategy. Conversely, customers acquired via brand ambassadors or direct search exhibit higher baseline brand trust. With an observed sample size of $N = 1,000$ customers (~250 per channel), the gap between TikTok's 21.79% retention rate and

Ambassador's 54.83% is wide enough to prove a structural difference in purchasing behavior rather than statistical noise.

### 2.2 CAC vs. LTV per channel

Using the cleaned dataset, three distinct formulas were applied to bridge gross sales and net profit per customer: LTV = total channel sales / number of unique channel customers; Gross Margin = LTV × 65%; Net Margin = Gross Margin − Estimated CAC of the channel. The consolidated result:

| Canal | Retention | LTV avg. | Gross margin | Estimated CAC | Net margin / customer |
|---|---|---|---|---|---|
| Ambassadors | 54.83% | US$84.29 | $54.79 | $8.00 | $46.79 |
| Organic | 52.94% | $83.38 | $54.20 | $0.00 | $54.20 |
| Instagram | 30.48% | MX$62.42 | $40.57 | $12.00 | US$28.57 |
| TikTok | 21.79% | $54.32 | $35.31 | $15.00 | $20.31 |

The analysis reveals a clear channel profitability hierarchy:

- **Organic (Highest Efficiency)**: Leads net profitability with $54.20 USD per customer due to zero customer acquisition cost ($0.00 CAC).
- **Ambassador (Top Paid Channel):** Leads paid/affiliate channels with an LTV of $84.29 USD and a net margin of $46.79 USD per customer.
- **Instagram (Moderate Performance):** Operates at a healthy equilibrium with $62.42 USD LTV and $28.57 USD net margin (deducting $12.00 USD CAC).
- **TikTok (Lowest Return)**: Ranks lowest with an LTV of $54.32 USD. Factoring in its $15.00 USD CAC, net profit drops to $20.31 USD per customer. The unit economic model is fragile: if TikTok's actual CAC exceeds its $35.31 USD gross margin threshold, the channel becomes immediately unviable

Note: Channel CAC benchmarks ($0 Organic, $8 Ambassador, $12 Instagram, $15 TikTok) and the 65% gross margin baseline were established as simulated industry benchmarks for D2C sports nutrition in Latin America. They were specified deliberately to maintain analytical rigor and financial sustainability—they were not extracted directly from the transactional dataset.

## 3. Data Pipeline Architecture

This dataset is synthetic by design, built in Python using Faker and Pandas to simulate 1,000 unique customers and 1,807 transactions over a 12-month period. The script assigned acquisition channels via a uniform random distribution (~25% initial volume per channel). Subsequently, asymmetric retention probabilities (65% Ambassador/Organic, 40% Instagram, 25% TikTok) were encoded exclusively into the retention probability variable to govern repeat purchases, decoupling initial acquisition from loyalty behavior.

### 3.1 Category Depletion Cycles

Each product category carries a distinct consumption cycle governing plausible repeat purchase timing:

| Product | Price | Depletion Cycle |
|---|---|---|
| Whey Protein | $55 USD | 30 days |
| Pre-Workout | $40 USD | 45 days |
| Creatine | $35 USD | 60 days |
| Multivitamin | $25 USD | 90 days |

On the base cycle of each product, a variability of [-5 to +15 days] is introduced before creating the next purchase event. The result of the pipeline is fitbrand_transactions.csv, which serves as raw input data for modeling in SQL and Power BI."

## 4. Modeling and Analytical Queries in SQL

Historical transaction logs were ingested into a relational SQLite database (fitbrand.db), transactions table, leveraging two key queries: a conditional aggregation query for channel retention and LTV, and a window function query to estimate repurchase velocity.

### 4.1 Retention and LTV per channel (conditional aggregation)

The query uses COUNT(DISTINCT ...) and CASE WHEN over purchase_number to isolate retained clients without the need for an auxiliary table:

```
SELECT
    acquisition_channel,
    COUNT(DISTINCT client_id) AS total_clients,
    COUNT(DISTINCT CASE WHEN purchase_number > 1 THEN client_id END) AS retained_clients,
    ROUND(
        CAST(COUNT(DISTINCT CASE WHEN purchase_number > 1 THEN client_id END) AS FLOAT) /
        COUNT(DISTINCT client_id) * 100, 2
    ) AS retention_rate_pct,
    ROUND(CAST(SUM(purchase_amount_usd) AS FLOAT) / COUNT(DISTINCT client_id), 2) AS avg_ltv_usd
FROM transactions
GROUP BY acquisition_channel
ORDER BY avg_ltv_usd DESC;
```

This query calculates overall cumulative dataset retention by evaluating customers with more than one transaction (purchase_number > 1), consolidating average LTV and historical repeat purchase rates for each acquisition channel.

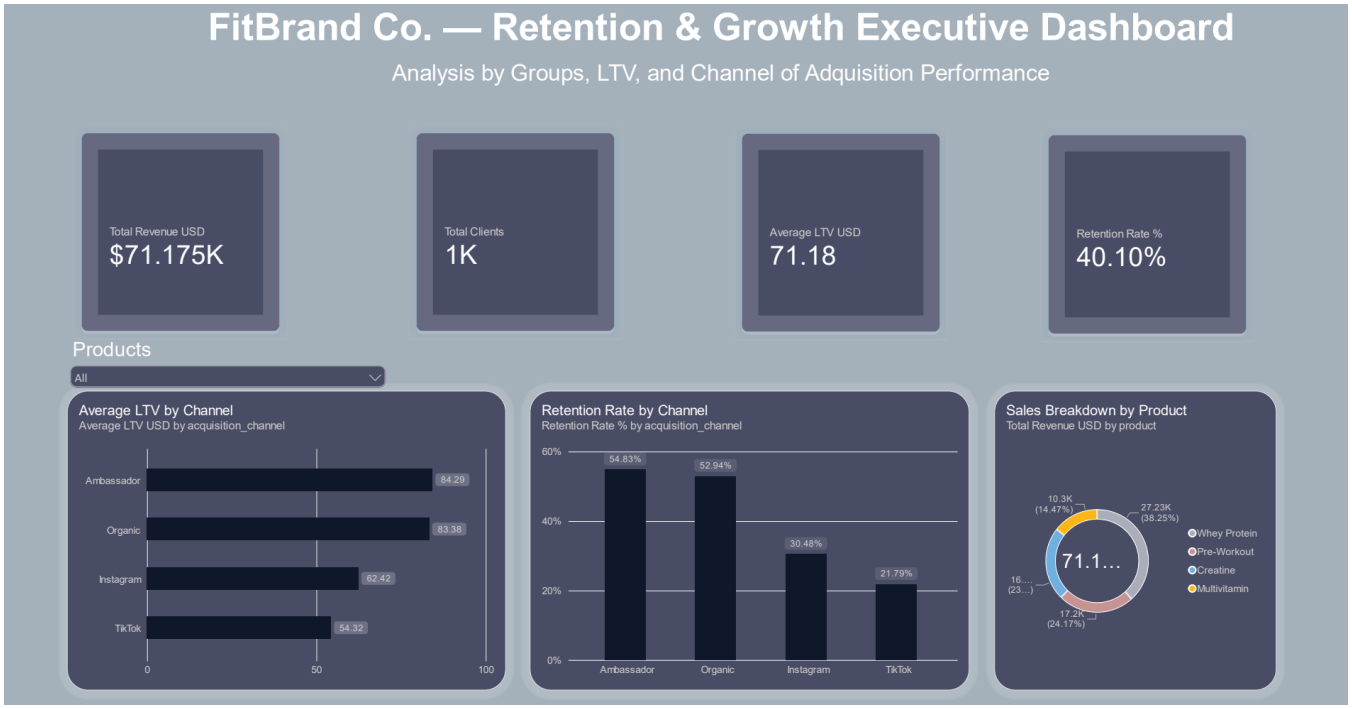
### 4.2 Repurchase Velocity via LAG()

To measure elapsed days between the initial purchase (purchase_number = 1) and first replenishment (purchase_number = 2) without correlated subqueries, transactions are partitioned by client and ordered chronologically:

```
WITH ordered_purchases AS (
  SELECT
    client_id,
    product,
    purchase_date,
    LAYER(purchase_date) OVER (
      PARTITION BY client_id
      ORDER BY purchase_number
    ) AS previous_purchase_date
  FROM transactions
)
SELECT
  product,
  ROUND(AVG(JULIANDAY(purchase_date) - JULIANDAY(previous_purchase_date)), 0) AS avg_days_to_repurchase
FROM ordered_purchases
WHERE previous_purchase_date IS NOT NULL
GROUP BY product;
```

This aggregation validates retargeting campaign timing: by grouping via window functions, SQL empirically extracts the average actual days customers take to replenish each product (35, 51, 65, and 95 days), establishing an empirical baseline to trigger ad spend right before or during product depletion.

## 5. Visual Diagnostics in Power BI




The relational model in Power BI utilizes transactions as the fact table, with acquisition_channel and purchase_number as core analytical dimensions. Four DAX measures were constructed on this table:

```
Total Revenue USD = SUM(transactions[purchase_amount_usd])

Total Clients = DISTINCTCOUNT(transactions[client_id])

Average LTV USD = DIVIDE([Total Revenue USD], [Total Clients], 0)

Retention Rate % =
VAR RetainedClients =
  CALCULATE(
    DISTINCTCOUNT(transactions[client_id]),
    transactions[purchase_number] > 1
  )
RETURN
  DIVIDE(RetainedClients, [Total Clients], 0)
```

The Executive Dashboard combines KPI cards (Total Revenue USD, Total Clients, Average LTV USD, Retention Rate %) with channel breakdowns. The visual layout confirms the SQL query output: TikTok's retention bar is noticeably shorter than Ambassador's, and channel average LTV mirrors the same trend, establishing the dashboard as a monthly tracking tool to verify whether budget reallocation is narrowing the gap.

## 6. Omnichannel Action Plan and Strategy

### 6.1 Budget reassignment: -30% TikTok, +Embajadores

**Reduce direct paid ad spend on TikTok by 30% and shift that capital into the Ambassador commission program**. A customer acquired via Ambassador generates, on average, $29.97 USD more in overall blended LTV than a TikTok customer. However, isolating retained customers reveals that a loyal Ambassador user achieves an LTV of ~$123.91 USD compared to TikTok's ~$111.08 USD (a marginal difference of only ~$12.83 USD). This demonstrates that TikTok does not underperform due to a low-value loyal segment, but strictly because of massive early cohort churn.

**Variable Commission Increase.** Raise code-attributed commission from 10% to 15%, incentivizing top-converting creators to post about the brand more frequently

**Seeding Strategy and Courtesy Kits.** Allocate a monthly budget to send free product packages (protein + creatine) to key athletes and micro-influencers to maintain continuous organic presence without paying cold ad spend.

**Maintaining the budget on Instagram**: Keep current Instagram spend unchanged. Generating a positive net margin of $28.57 USD per customer, it serves as a stable mid-tier volume channel while migrating capital from TikTok to Ambassador.

### 6.2 Educational Retargeting Segmented by Consumption Cycle (CEO/Athlete)

Structure Custom Audiences in TikTok Ads Manager targeting first-time buyers (purchase_number = 1). Ad delivery is not executed en masse, but through automated triggers synchronized with each SKU's depletion cycle:

**Whey Protein (30-day base):** Retargeting Trigger on Day 35.

**Pre-Workout (45-day base):** Retargeting Trigger on Day 50.

**Creatine (60-day base):** Retargeting Trigger on Day 65.

**Multivitamin (90-day base):** Retargeting Trigger on Day 95.

Ad creative avoids aggressive sales pitches, featuring organic educational videos from the CEO or brand athletes ("3 Common Mistakes Diluting Your Protein Results") with a direct replenishment link. Targeting a narrow, high-intent audience requires minimal ad spend (a micro-budget of $200–$300 USD/month in retargeting CPMs). Converting just an additional 10% of TikTok's cohort generates over $2,400 USD in gross revenue while maintaining a highly profitable Return on Ad Spend (ROAS).

### 6.3 Checkout Cross-Selling

When a TikTok user purchases Pre-Workout ($40 USD), checkout should natively display a dynamic bundle with Creatine or Whey Protein at an optimized price point. This addresses the issue at its root during the first transaction, rather than relying exclusively on win-back campaigns after the customer has already churned.

### 6.4 Loyalty and Subscription Program (Retention Levers):

Implement a recurring purchase model (Subscribe & Save) with an automated 10% discount on the second delivery, combined with a points-based loyalty system redeemable exclusively on the second replenishment order.